No Arbitrage and the Existence of a Pricing Measure: A Machine-Checked Proof of the Discrete Fundamental Theorem of Asset Pricing

ftap-proofs: eigenq-xyz research series

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The Fundamental Theorem of Asset Pricing states that a finite-state, discrete-time market admits no arbitrage if and only if there exists an equivalent martingale measure under which discounted asset prices are martingales. We give a complete machine-checked proof of the discrete Harrison-Pliska theorem in Lean 4, verified against mathlib with zero proof gaps. The forward direction is a short separation argument; the reverse direction is the substantive content, established through a state-price functional obtained by a separating-hyperplane argument on the cone of attainable payoffs. The development is structured so that the market model, the no-arbitrage condition, and the martingale-measure condition are reusable by downstream pricing results. As an empirical companion we run a controlled study of the theorem's central prediction: put-call parity, a model-free consequence of a single consistent pricing measure, should hold across strikes. On an arbitrage-free Black-Scholes cross-section perturbed by realistic quote friction, the parity deviation stays under one basis point of spot when quotes are tight and widens to roughly 48 basis points on average, and 120 at the 95th percentile, under stressed quotes in a high-volatility regime. The generator is synthetic, so this is a controlled demonstration of the methodology rather than a claim about a specific market, and the same harness runs on real quotes unchanged. The contribution is a verified foundation for derivative pricing, not a trading signal: the proof guarantees the theorem as stated, and the study measures how a frictionless prediction degrades as friction grows.

Contents

1 Introduction

Does the absence of arbitrage in a market guarantee that a single, consistent set of prices explains every traded asset? The Fundamental Theorem of Asset Pricing answers yes, and this paper makes that answer machine-checked.

The practitioner stakes are concrete. Every relative-value desk, every option model, and every risk system rests on the assumption that prices admit a risk-neutral measure. When that assumption is stated as folklore rather than as a theorem with explicit hypotheses, it is easy to apply it where its assumptions fail, in markets with frictions, in incomplete markets, or over horizons where the discrete model is the wrong idealization. A proof states exactly what must hold for the conclusion to follow.

We formalize the discrete Harrison-Pliska theorem in Lean 4. On a finite probability space with a finite trading horizon, we prove that the market admits no arbitrage if and only if there exists a probability measure, equivalent to the reference measure, under which every discounted asset price is a martingale. The forward implication follows from the martingale property by a short argument. The reverse implication is the substantive one: from no arbitrage we construct a strictly positive linear pricing functional on the space of attainable payoffs by separating the no-arbitrage cone from the nonnegative orthant, then normalize that functional into the desired measure.

The contribution is threefold. First, the theorem is established with no admitted steps and no axioms beyond those of mathlib, so the result is checked rather than asserted. Second, the formal model is written as a reusable interface: a market, a trading strategy, a no-arbitrage predicate, and a martingale-measure predicate, so that subsequent pricing results cite the theorem rather than re-deriving it. Third, we pair the proof with an empirical exercise that asks how well the frictionless conclusion describes a real market, reporting the magnitude of the gap rather than only its statistical significance.

The remainder of the paper is organized as follows. Section 2 fixes the market model and states the assumptions. Section 3 states the theorem and sketches both directions. Section 4 documents the machine-checked artifacts and the precise scope of what is proven. Section 5 describes the empirical sample, Section 6 reports the pricing-measure consistency test, and Section 7 reports where it holds and where it breaks. Section 8 concludes.

2 Setup

The market is a finite, discrete-time, frictionless model. We fix the primitives so that the theorem statement in Section 3 is unambiguous, and we match the notation of the formal development where it does not conflict with standard usage.

Definition 2.1. (Market.) A market consists of a finite probability space, a finite trading horizon with dates indexed from the start to the terminal date, a finite collection of risky assets with adapted price processes, a strictly positive deterministic numeraire normalized to one at the start, and a filtration that encodes the information available at each date. The reference probability measure assigns positive mass to every state.

Discounting is by the numeraire: the discounted price of an asset is its price divided by the numeraire at that date. A trading strategy assigns, at each date, holdings in each risky asset and the numeraire, with holdings chosen on the information available at the previous date. A strategy is self-financing when each rebalancing is funded entirely by the current portfolio, with no inflow or outflow.

We state the standing assumptions as numbered displays rather than burying them in prose, because the theorem holds exactly under them and not otherwise.

> **Assumption A1 (Finite state space and horizon).** The set of states is finite, and trading occurs at finitely many dates.
>
> **Assumption A2 (Frictionless trading).** Trading is costless, positions are unconstrained in sign and size, and assets are perfectly divisible.
>
> **Assumption A3 (Positive numeraire).** The numeraire is strictly positive at every date and normalized to one at the start.

An *arbitrage* is a self-financing strategy with zero initial cost, nonnegative terminal value in every state, and strictly positive terminal value in at least one state. The market is *arbitrage-free* when no such strategy exists. A probability measure is an *equivalent martingale measure* when it agrees with the reference measure on which states have positive probability and every discounted asset price is a martingale under it.

3 Main Results

Theorem 3.1. ***(Fundamental Theorem of Asset Pricing, discrete case.)*** *Under Assumptions A1 through A3, the market is arbitrage-free if and only if there exists an equivalent martingale measure.*

Proof sketch. The reverse direction first. Suppose an equivalent martingale measure exists. The discounted value of any self-financing strategy is then a martingale under that measure, so its terminal expectation equals its initial value. A candidate arbitrage has zero initial cost, hence zero expected discounted terminal value, yet a nonnegative payoff that is strictly positive in some state of positive measure. A nonnegative random variable with zero mean under a measure of full support must be zero everywhere, contradicting the strict-positivity requirement. No arbitrage can exist.

The forward direction carries the content. Suppose the market is arbitrage-free. The set of discounted terminal payoffs attainable at zero cost is a linear subspace; the requirement of no arbitrage says this subspace meets the nonnegative orthant only at the origin. Separating the subspace from the compact base of the orthant by a hyperplane yields a strictly positive linear functional that annihilates every zero-cost attainable payoff. Representing that functional by its values on the states and normalizing gives a strictly positive probability vector, equivalent to the reference measure, under which discounted prices have constant conditional expectation, which is the martingale property. This is the desired measure.

Both directions are established formally; the precise statements and the machine-checked artifacts are documented in Section 4.

4 Formalization

This section is mandatory for every paper in the eigenq-xyz series. It records which mathematical claims are machine-checked, which are not, and where the artifacts live.

4.1 Lean 4 artifacts

The formal development lives in `ftap-proofs/`. The following table maps the paper-level statements to their Lean 4 names.

Paper statement	Lean 4 name	File
Theorem 3.1 (full)	`FtapProofs.ftap`	`ftap-proofs/FtapProofs/Theorem.lean`
Reverse direction (EMM ⟹ NA)	`FtapProofs.emm_implies_no_arbitrage`	`ftap-proofs/FtapProofs/Theorem.lean`
Forward direction (NA ⟹ EMM)	`FtapProofs.no_arbitrage_implies_emm`	`ftap-proofs/FtapProofs/Theorem.lean`
Value process is a martingale under any EMM	`FtapProofs.discountedValue_martingale_of_emm`	`ftap-proofs/FtapProofs/MartingaleMeasure.lean`
Risk-neutral pricing identity	`FtapProofs.risk_neutral_pricing`	`ftap-proofs/FtapProofs/MartingaleMeasure.lean`

Formalization claim

At the head of the `main` branch, `lake build` succeeds for `ftap-proofs` and a search for `sorry` across the proof tree returns no matches. Reproduce with the `verify-lean` skill.

4.2 Scope of formalization

- **Formalized.** The biconditional in Section 3 is proven from the assumptions in Section 2 with no axiom beyond mathlib. The separating-hyperplane construction, the martingale property of the discounted value process, and the risk-neutral pricing identity are all machine-checked.
- **Not formalized.** Everything in the empirical sections below. The data ingestion, the estimation of an implied pricing measure, and the deviation statistics are numerical and rely on a continuous model used as an approximation. They are not theorems and are not claimed to be.

5 Empirical Setup

The companion study measures the operational shadow of a consistent pricing measure: put-call parity. The theorem guarantees a single measure exists in a frictionless arbitrage-free market, and parity, $C - P = S - Ke^{-rT}$, must then hold at every strike independent of any volatility model. The study asks how large the parity deviation becomes once realistic quote friction is introduced, and how that depends on the volatility regime.

Committable real option histories are licensed and out of scope for a public repository, so the study uses a synthetic generator. This is a controlled demonstration, not a claim about a specific market. The generator is an arbitrage-free Black-Scholes cross-section, so parity holds exactly before friction is added; each quote is then perturbed by independent multiplicative Gaussian noise at four friction levels, from tight to stressed, and the grid is repeated across three volatility regimes. The same harness runs on a real option snapshot by replacing the generator with a loader.

Field	Value
Generator	Arbitrage-free Black-Scholes call and put cross-section
Spot, rate, maturity	$S = 100$, $r = 3\%$, $T = 0.25$ years
Strikes	80 to 120 in steps of 2.5 (17 strikes)
Friction	Multiplicative quote noise, std 5, 25, 100, 300 bps (tight to stressed)
Volatility	15%, 30%, 60% annualized (low, mid, high regime)
Trials	20,000 per cell, single seeded generator

The generating script and its committed output are `ftap-proofs/results/parity_consistency_study.py` and `ftap-proofs/results/metrics/parity_consistency.json`.

6 Results

Table 6.1 reports the mean parity deviation in basis points of spot for every volatility regime and friction level. The pattern is the one the theorem predicts. When quotes are tight the deviation is under one basis point on average, so the frictionless prediction essentially holds. As friction grows the deviation rises roughly in proportion to the quote noise, reaching tens of basis points under stressed quotes. The volatility regime amplifies the effect: parity is model-free and does not depend on volatility directly, but higher volatility raises option price levels, so a fixed relative quote noise produces a larger absolute parity gap.

Table 6.1: Mean put-call parity deviation in basis points of spot, by volatility regime and quote-friction level. Source: `ftap-proofs/results/metrics/parity_consistency.json`, 20,000 trials per cell.

Volatility regime	Tight	Normal	Wide	Stressed
Low (15%)	0.45	2.26	9.08	27.16
Mid (30%)	0.54	2.70	10.81	32.37
High (60%)	0.80	3.99	15.94	47.86

The tail behaves the same way. At the 95th percentile across strikes and trials, the deviation runs from about 1.3 basis points in the tight, low-volatility cell to about 119 basis points in the stressed, high-volatility cell.

7 Robustness

The controlled design lets us read robustness directly off the grid rather than re-estimate on subsamples.

Friction sensitivity. Across all three volatility regimes the mean deviation rises monotonically with quote noise, and close to linearly: each step up in friction produces a proportional step up in deviation. There is no friction level at which the parity relation silently breaks; it degrades smoothly, which is the behavior a no-arbitrage diagnostic should have.

Regime conditioning. Moving from the low to the high volatility regime raises the stressed-quote mean deviation from 27 to 48 basis points, an increase of about three-quarters, and the worst cell reaches a 95th-percentile deviation near 119 basis points. A desk reading parity deviations as a no-arbitrage signal would therefore need a regime-aware threshold: a deviation that flags mispricing in calm conditions sits inside the friction band on a stressed, high-volatility tape.

Quote-convention sensitivity. The study perturbs both legs with independent noise, the conservative case. Common-mode quote movement, such as a shared shift from a stale underlier, cancels in the parity difference and would reduce the measured deviation. The reported numbers are therefore an upper bound on the gap attributable to idiosyncratic quote noise.

Where it would break on real data. The synthetic generator omits early-exercise premia, dividend timing, and borrow costs, each of which moves parity in a direction the frictionless relation does not capture. On real quotes these would add a structural deviation on top of the friction band measured here, so the study's numbers should be read as the friction-only floor.

8 Discussion

The proof and the planned measurement answer two different questions. The proof settles a mathematical equivalence: in a frictionless finite market, no arbitrage is exactly the existence of a consistent pricing measure, with no hidden assumptions. The measurement asks how good an approximation that idealization is for a market with frictions, and the honest expectation is that the gap is small in calm regimes and widens in stress. Neither result is a trading signal. The value of the verified foundation is that every downstream pricing claim in this series can cite a checked theorem rather than restate folklore, and can be explicit about the assumptions under which its prices are guaranteed to be consistent.

Three extensions are natural. The first is the multi-period incomplete-market case, where the pricing measure is no longer unique and the object of interest is the set of consistent measures. The second is an explicit treatment of transaction costs, where no arbitrage corresponds to a cone of consistent price systems rather than a single measure. The third is the empirical exercise above, executed across several underliers and reported with the deviations broken out by regime.

9 Appendix

9.1 A. Proof references

The full proofs are machine-checked; rather than reproduce them, we point to the source. The forward direction's separation argument is in `ftap-proofs/FtapProofs/Theorem.lean`, building on the attainable-payoff and no-arbitrage definitions in `ftap-proofs/FtapProofs/Arbitrage.lean`. The martingale machinery is in `ftap-proofs/FtapProofs/MartingaleMeasure.lean`.

9.2 B. Replication

```
git clone https://github.com/eigenq-xyz/quant-proofs
cd quant-proofs/ftap-proofs
lake build              # machine-checks the proof; zero sorry on main

# Reproduce the controlled parity study (Section 7) with a numpy/scipy
 ↪  environment:
python results/parity_consistency_study.py   # writes
 ↪  results/metrics/parity_consistency.json
```

The study is seeded, so the committed `parity_consistency.json` reproduces exactly. Table 6.1 is read directly from that file.